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Vagueness and the trade-off between the classification and delineation of geographic regions – an ontological analysis

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For inherently vague and granular phenomena such as ecoregions, ecosystems, biomes, and biotopes, the interplay of granularity and vagueness leads to a trade-off in the classification and delineation of such phenomena: the goal of preciseness (lack of vagueness) of the delineation contradicts the goal of building a sophisticated classification system using the Aristotelian method of classification. This trade-off is based on the reliance on *local* qualities for a precise delineation of particular regions and the reliance on *nonlocal* qualities that serve as differentia in the Aristotelian classification. An ontological analysis of the logical interrelations between vagueness, granularity, and scale is critical for developing logically rigorous, nonlocal, and nonarbitrary classification and delineation systems for inherently vague and granular geographic phenomena.

Keywords: applied ontology; mereology; vagueness; granularity; top-level ontology; geospatial ontology; ecoregion classification and delineation

1. Introduction

Classification and delineation systems of geographic regions are important tools for analyzing and representing geographic phenomena. Map-based representations of such systems include what Bunge (1962) calls Area-Class maps and what Chrisman (1982, 1997) calls Categorical Coverages. Building classification and delineation systems for inherently vague and granular geographic regions such as ecoregions, climate zones, biomes, and ecozones, etc. that are objective, logically rigorous, and nonlocal has proven to be difficult (Bowker 2000a,b, McMahon *et al.* 2001, Waterton 2002, Loveland and Merchant 2004). Many experts do not seem to be able to agree on the ontological status of certain categories of geographic regions. Opinions range from the belief that many kinds of geographic regions exist in reality and can be identified using scientific methods of classification and delineation (e.g., Bailey 1983), to the belief that the classification and delineation of geographic regions is more or less arbitrary or mostly policy-based (e.g., Bowker 2000a,b).

To help clarifying the ontological status of inherently granular and vague geographic regions, as well as the logic of their classification and delineation, three important aspects of such georegions are analyzed from an ontological perspective: (i) the granular and scale-dependent nature of the relation between qualities such as landform, vegetation, and so on, and the geographic regions in which they inhere; (ii) the vagueness affecting the

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classification of geographic regions based on such qualities; and (iii) the vagueness affecting the location of boundaries that separate regions with different qualities. The interplay of granularity and vagueness indicated in (i–iii) leads to a fundamental trade-off between the classification and the delineation of such geographic regions:

Conjecture 1: For geographic regions that are subject to granularity and vagueness in the sense of (i–iii), it is impossible to have both: a precise (non-vague) de-lineation and a sophisticated Aristotelian classification based on genus, species and quality-based differentia.

This trade-off is due to the reliance on *local* qualities for a (relatively) precise delineation on one hand and the reliance on *nonlocal* qualities for classification purposes on the other. In the presented ontology it will be possible to support these claims by a semiformalized argument.

2. Related work

Vagueness, granularity, and boundary indeterminacy are widely discussed in the literature on the classification and delineation of geographic regions, including Chrisman (1982), Burrough and Frank (1995), Ehlschlaeger and Goodchild (1994), Goodchild (2000), and Hunsaker *et al.* (2001). Existing approaches for dealing with those problems are mainly based on set theory (e.g., functions mapping coordinates or sets of coordinates to attribute values) and have been used successfully for modeling and representation purposes (e.g., Goodchild 2003).

Unfortunately, set theory is less useful as a basis for formal ontology (Smith 1994, Casati *et al.* 1998, Varzi 2001). This is reflected by the fact that the most widely used formal ontologies, including DOLCE (Gangemi *et al.* 2003), BFO (Grenon and Smith 2004), and the Open Biological Ontology (OBO) (obofoundry.org 2010), are all based on mereology, a theory of parts and wholes (Simons 1987), rather than on set theory. In addition, although there are many attempts to *model* and to *represent* vagueness and indeterminacy based on fuzzy set theory (Zadeh 1975) and/or set theory extended by probabilistic and stochastic tools, set theory does not seem to be the best tool to formalize the *logic* of vagueness (Elkan 1994, Keefe and Smith 1996, Bennett 1998).

In this article, a mereology-based theory of the classification and delineation of geographic regions is presented. This theory extends the top-level ontology of Bittner *et al.* (2009). The theory can be seen as a mereology-based counterpart to set-theory-based approaches to formalize the ideas underlying Categorical Coverages and Area-Class maps. The underlying theory of vagueness is based on supervaluation (Fine 1975, Keefe and Pinkal 1995, Smith 1996, Bennett 1998). The formal treatment of granularity and scale extends work on the Order of Magnitude reasoning in Artificial Intelligence (Raiman 1991, Dague 1993a,b).

To illustrate the formal presentation, two ecoregion classification and delineation systems are used as running examples: the ecoregion system of Bailey (1983) promulgated by the USDA Forest Service (1992), USFS (1995), and the system of Omernik (1987, 1995, 2004) promulgated by the US Environment Protection Agency (CEC 1997). The two systems are used as examples because they attempt to delineate similar kinds of geographic regions – ecoregions – but they approach the trade-off between the classification and delineation of geographic regions (as stated in (1)) in different ways.

The discussion of Bailey's system extends existing work by Sorokine and Bittner (2005), Sorokine *et al.* (2006), and Bittner (2007) in which the relations between the taxonomic structure and the partonomic structure in Bailey's system are analyzed. However, Sorokine

and Bittner (2005) and Sorokine *et al.* (2006) considered only the special case in which every class/universal has exactly one instance. Bittner (2007) formalized partonomic and taxonomic structures of Bailey's system in a general mereology-based theory. This theory is extended in this study by a formalization of the logical interrelations between the partonomic and taxonomic structures based on the notion of homogeneity of geographic regions with respect to certain geographic qualities. In addition, the formal analysis here takes into account granularity, scale-dependency, and vagueness. None of those aspects have been addressed in the previous work.

GIScientists such as Kronenfeld (2005) addressed some aspects of the trade-off between the classification and delineation of geographic regions (1) from representational and cartographic perspectives. Relationships between this body of work and the presented formal ontology-based approach are discussed in Section 6.2.

3. Review of relevant top-level predicates

Top-level ontologies specify the semantics for general unary predicates (or sorts) such as universal, particular, as well as binary (and n -ary) predicates such as instance-of, i-part-of (short form for instance-level-part-of), universal-part-of, sub-universal-of, exactly-the-same-size, roughly-the-same-size, and negligible-in-size, (Gangemi *et al.* 2003, Agarwal 2005, Bittner and Donnelly 2006, Bittner 2009, Bittner *et al.* 2009).

3.1. Mereotopology, universals, and qualities

Let the particulars YNP_M , YNP_I , and YNP_W be the i-parts-of the particular Yellowstone National Park (YNP) that are occupied by the particulars Montana, Idaho, and Wyoming respectively. Then YNP is the sum of YNP_M , YNP_I , and YNP_W . Moreover, YNP is partitioned by the collection $\{YNP_M, YNP_I, YNP_W\}$. YNP is self-connected. By contrast, the territory of the United States is not self-connected.

The particular YNP is an instance of the universal *National Park*. The specific size of YNP is a quality particular and inheres in YNP. It is an instance of the quality universal *Size*. The particular instance of the quality universal *Size* that inheres in YNP can be characterized mathematically using the Lebesgue measure Burk (1997) of the region that is occupied by YNP. The Lebesgue measure of the region of YNP is (currently) 3472 square miles. The quality universal *Size* is a sub-universal-of the universal *Quality*. Generally, universals are understood here as types in the sense of Guarino and Welty (2000). That is, universals are structured by the relation sub-universal-of into tree-like structures.

Consider the three maps of ecoregions of North America in Figure 1 CEC (1997). Each map represents a collection of ecoregion particulars. The symbols *LI*, *LII*, and *LIII* are used to name the collections that the members are depicted on the corresponding maps. The collection *LI* partitions the continent of North America (*NAC*): the members of *LI* jointly sum up to *NAC* and no two distinct members of *LI* overlap. Analogously, the collection *LII* is a more fine-grained partition of *NAC* and the collection *LIII* is an even more fine-grained partition of the contiguous USA (*CUS*). The collections *LI*, *LII*, and *LIII* not only partition *NAC/CUS*, but they are also hierarchically nested or partonomically included ($\uparrow$ -p-included) within one another. That is, every member-of *LII* is an i-part-of some member-of *LI*. Analogously, *LIII* is $\uparrow$ -p-included in *LII*, and *LIII* is $\uparrow$ -p-included in *LI*.

Corresponding to partonomic inclusion predicates there are universal parthood predicates. The universal *County* is an upward universal-part-of ($\uparrow$ -u-part-of) the universal

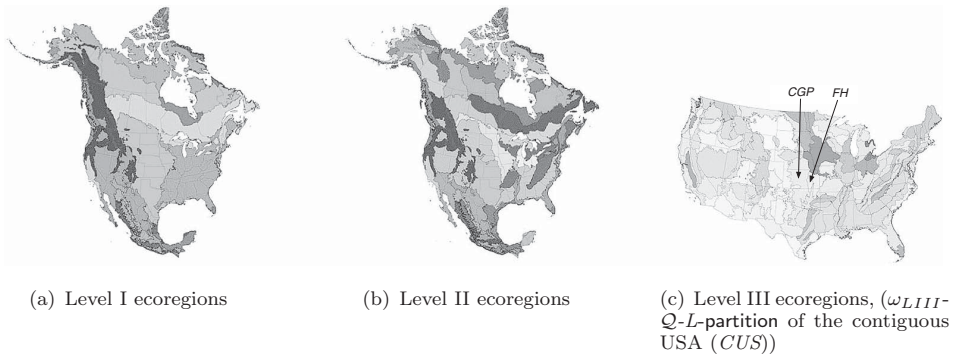


Figure 1. Ecoregions of North America at different scales (CEC 1997).

State: every instance of *County* is an i-part-of some instance of *State* (see Bittner *et al.* 2009 for details).

3.2. Qualitative size predicates

Top-level predicates that refer to qualitative size relations are critical for formalizing how granularity, scale-dependency, and vagueness affect the classification and delineation of geographic regions. Here the binary predicates exactly-the-same-size, roughly-the-same-size, negligible-in-size, and same-scale are introduced. The intended meaning of these predicates is specified by simple equations and constraints (Raiman 1991, Dague 1993a,b, Bittner and Donnelly 2006, Bittner 2009).

Let $\|x\|$ and $\|y\|$ be functions that yield the Lebesgue measures (Burk 1997) (i.e., the exact sizes) of the regions x and y . Let ω be a parameter that lies within a range of $[\omega_{\downarrow}, \omega_{\uparrow}]$ such that $0 < \omega_{\downarrow} \leq \omega \leq \omega_{\uparrow} \leq 0.6$.

Definition 3.1: (i) Regions x and y have exactly-the-same-size if and only if $\|x\| = \|y\|$; (ii) Regions x and y have roughly-the-same-size on the precisification ω (x and y have ω -roughly-the-same-size) if and only if $1/(1 + \omega) \leq \|x\| / \|y\| \leq 1 + \omega$; (iii) Region x is negligible-in-size relative to y on the precisification ω (x is ω -negligible-in-size relative to y) if and only if $\|x\| / \|y\|$ is smaller than $\omega/(1 + \omega)$; and (iv) Regions x and y are of the same-scale on the precisification ω (x and y are of ω -same-scale) if and only if $\omega/(1 + \omega) \leq \|x\| / \|y\| \leq (1 + \omega)/\omega$.

The choice of the particular class of constraints in Definitions 3.1(ii–iv) is to a certain degree arbitrary. Other classes of constraints and different ranges of parameters can be used if they satisfy the Axioms (A8–A27) of (Bittner 2010). This particular class of constraints used here is simple and has proven useful in practice (Mavrovouniotis and Stephanopoulos 1988, Feng and Bittner to appear).

Definition 3.1 clarifies important semantic differences between the *crisp* predicate exactly-the-same-size and the *vague* predicates roughly-the-same-size, negligible-in-size, and same-scale. The semantics of the crisp predicate is specified in Definition 3.1(i) using a *parameter-less equation*. This formally expresses that there is a unique and determinate way of interpreting the predicate exactly-the-same-size.

By contrast, the predicates roughly-the-same-size, negligible-in-size, and same-scale are *vague*. According to the semantic understanding of vagueness (Fine 1975, Keefe and Pinkal

1995, Smith 1996), there is no unique and completely determinate way of interpreting vague predicates. In Definition 3.1(ii–iv) this is reflected formally by *parameterized constraints*. The parameter ω is called a precisification parameter. A particular choice of ω represents one particular way of making the meaning of the vague predicates roughly-the-same-size, negligible-in-size, and same-scale precise. For a particular choice of the precisification parameter ω , the constraints in Definition 3.1(ii–iv) hold for certain ranges of the ratio $\|x\| / \|y\|$.

It follows from Definition 3.1:

Lemma 3.2: *If x and y are of ω_1 -roughly-the-same-size and $0 < \omega_1 \leq \omega_2 \leq 0.6$, then x and y are of ω_2 -roughly-the-same-size. Analogously for negligible-in-size and same-scale.*

3.3. Admissible ranges of precisifications

The predicates roughly-the-same-size, negligible-in-size, and same-scale admit certain ranges $\Omega = [\omega_{\downarrow}, \omega_{\uparrow}]$ of precisification parameters. For a given choice of $\omega_{\downarrow}$ and $\omega_{\uparrow}$, all precisifications $\omega \in [\omega_{\downarrow}, \omega_{\uparrow}]$ are semantically indistinguishable, that is, are semantically equivalent ways of making the meaning of the vague predicates roughly-the-same-size, negligible-in-size, and same-scale precise.

The following conventions are adopted to distinguish crisp and vague predicates: All vague predicates have either ω or Ω as a prefix. Crisp predicates do not have prefixes. The prefixes ω and Ω can be thought of as semantic operators that specify a given vague predicate is true on the precisification ω or true for the range of precisifications Ω .

To determine admissible ranges of the parameter ω (i.e., the values for $\omega_{\downarrow}$ and $\omega_{\uparrow}$ that capture the meaning of a given vague predicate) or to fix those ranges by *fiat* is a scientific and empirical question and not a matter of ontology or logic (Bittner and Donnelly 2006, Bittner 2010). Therefore, it is not a part of the ontological analysis presented here. An example of how to fix admissible ranges of precisifications in a specific scientific context is presented in Feng and Bittner (to appear).

4. Qualities, quality pattern, and homogeneity

The top-level predicates reviewed in Section 3 provide the formal basis for a mereology-based formal theory of classification and delineation systems for geographic regions. Central to such a theory are the notions of quality universal, quality pattern, and homogeneity with respect to quality pattern, which are introduced in this section.

4.1. Environmental qualities for the classification of geographic regions

To illustrate the definitions of quality pattern and homogeneity and to support the presentation of the running examples, two groups of quality universals are used: *Land-surface* quality universals and *Climate* quality universals.

Land-surface quality universals characterize important aspects of the surface of geographic regions. The quality universal *Land-surface quality* has at least four sub-universals (Omernik 1987, Table 1): *Land-surface form*, potential natural vegetation (*Climax Vegetation*), *Land use*, and *Soil type* (Table 1). (Other Land-surface quality universals include *Geomorphology*, *Lithology*, and *Stratigraphy*, etc. (USDA Forest Service 1992), but are omitted here.)

Climate quality universals fall into at least two major groups (Bailey 1983): climate types and climate regimes. The sub-universals of the quality universal *Climate type* are

Table 1. Land-surface qualities of geographic regions – Central Great Plains and Flint Hills as examples (Omernik 1987).

Geographic region	Land-surface form (LF)	Climax vegetation (CV)	Land use (LU)	Soil type (S)
...	...	...	...	...
Central great plains	Irregular plains	Bluestem/grama prairie, bluesteam prairie, buffalo grass	Cropland, cropland with grazing land, some irrigated agriculture	Dry mollisols
Flint hills	Open Hills	Bluestem prairie (Bluestem, panic indian grass) (Hapludolls)	Subhumid grassland and semiarid grazing land	Mollisols
...	...	...	...	...

distinguished based on the classification of the weather averaged over a long time period (American Meteorological Society 2007). The sub-universals of *Climate regime* are identified through the classification of the seasonal fluxes of energy and moisture and the resulting broad climatic similarities and (climax) vegetation (Koeppen 1931, Bailey 1983). Examples include the quality universals *Humid Temperate Climate Regime*, *Prairie Climate Type*, and so on.

4.2. Scale, granularity, and quality pattern

Quality universals such as *Land-surface form*, *Climax Vegetation*, *Land use*, and *Climate type*, are instantiated by quality particulars that inhere in specific regions of the surface of the Earth. To formally specify how quality universals are the basis for the classification and delineation of ecoregions and other kinds of geographic regions, three important aspects need to be taken into account:

First, the relation between quality universals such as *Land-surface quality* and *Climate quality* and regions of the surface of the Earth is *scale dependent*. Consider the geographic region that is called Central Great Plains in the system of the EPA. (The Great Plains are frequently used in the literature as examples of how vagueness affects geographic regions, (e.g., Rossum and Lavin 2000).) In Figure 1c this geographic region is labeled *CGP*. According to Omernik (1987, Table 1) this region is characterized by the land-surface form quality universal *Irregular plain* (cf. Table 1). That is, quality particulars that instantiate the quality universal *Irregular plain* characterize the region called Central Great Plains. Clearly, the quality universal *Irregular plain* does not have an instance that characterizes a region on the surface of the Earth that is smaller than a region of geographic scale. Similarly, *Irregular plain* does not have an instance that characterizes a geographic region of continental scale. Thus, *Irregular plain* has only qualities as instances that characterize regions of certain *scales*.

The lower scale bound is necessary whereas the upper bound is contingent. That is, instances of quality universals such as *Irregular plain* can (by definition) inhere only in regions of geographic scale or larger. By contrast, it is just accidentally (in the ontological sense) the case that currently the quality universal *Irregular plain* does not have an instance that inheres in a geographic region of continental scale. This might have been different in the past, and this may be different in the future.

Definition 4.1: A quality universal Q is a sub-universal of *Geographic quality* if and only if all instances of Q inhere in regions of geographic scale or larger.

For example, *Land-surface quality* and *Climate quality* are geographic qualities. By contrast, *Size* is not a geographic quality because it also has instances that inhere in smaller regions.

Second, the relation between quality universals and geographic regions is *granular*. For example, to say that the geographic region Central Great Plains is characterized by the land-surface form quality universal *Irregular plain* does not mean that every i-part of geographic scale of this region has a land-surface form which is an instance of *Irregular plain*. There may be comparatively small (geographic-scale) regions in which different land-surface forms are instantiated. Analogously for climate types: There may, for example, exist comparatively small (geographic-scale) regions near larger water bodies where the average temperature (and thus the climate type) is different from the larger surroundings.

Third, geographic regions are characterized not only by single quality universals but also by sets or pattern of quality universals. For example, the geographic region Central Great Plains is (among others) characterized by sub-universals of the quality universals *Land-surface form*, *Climax Vegetation*, *Land use*, and *Soil type* (Table 1, row 3).

4.3. Homogeneity and quality pattern

To establish a formal relationship between particular geographic regions and pattern of quality universals that accounts for scale and granularity, the notion of homogeneity with respect to a quality pattern is introduced.

Let $Q_1, \dots, Q_n$ be variables ranging over geographic quality universals such as *Climate Type*, *Climate Regime*, *Climax Plant Formation*, *Land-surface form*, *Land use*, and *Soil type*, as well as sub-universals of such qualities.

Definition 4.2: A geographic region x is ω -homogeneous with respect to the quality pattern $\mathcal{Q} = \langle Q_1, \dots, Q_n \rangle$ (ω - $\mathcal{Q}$ -homogeneous) if and only if for every j such that $1 \leq j \leq n$: the sum of all the i-parts of x , which are the regions of geographic scale and in which inheres an instance of Q_j , has ω -roughly-the-same-size as x . In addition, every quality universal occurs at most once in a quality pattern. That is, if $Q_j = Q_k$ then $j = k$.

The quality pattern $\mathcal{Q}$ characterizes the region x on the precisification parameter ω if and only if x is ω - $\mathcal{Q}$ -homogeneous. The homogeneity predicate is vague because it is defined in terms of the vague predicate roughly-the-same-size.

Consider the geographic region labeled *CGP* (Central Great Plains) in Figure 1c. *CGP* is ω -homogeneous with respect to a quality pattern that is formed by the quality universals listed in row 3 of Table 1. Let CGP^{lf} , CGP^{cv} , CGP^{lu} , and CGP^s be the sums of i-parts of *CGP* at which sub-universals of the quality universals *LF*, *CV*, *LU*, and *S*, respectively, are instantiated as listed in the third row of the table. These i-parts of *CGP* have the following properties: First, in CGP^{lf} inheres the quality particular *lf*. This *lf* is an instance of the quality universal *Irregular plain* (column 2, row 3 of Table 1). *Irregular plain*, in turn, is a sub-universal of *Land-form* (*LF*, column 2, row 1 of Table 1). Analogously for CGP^{cv} , CGP^{lu} , and CGP^s . Second, CGP^{lf} , CGP^{cv} , CGP^{lu} , and CGP^s are not necessarily self-connected i-parts of *CGP*. All overlap to a significant degree and have ω -roughly-the-same-size as *CGP*.

This example illustrates that, according to Definition 4.2, a ω - $\mathcal{Q}$ -homogeneous geographic region may have minor i-parts of geographic scale in which a certain quality universal Q_Q of $\mathcal{Q}$ is not instantiated (Q_Q -inhomogeneities). The definition, therefore,

accounts for *granularity* in the sense that spatial variations of quality pattern in geographic regions below a certain size threshold do not affect the characterization of the whole as a ω - Q -homogeneous region (they are, in effect, ignored). The size threshold is such that the sum of Q_Q -inhomogeneities of a ω - Q -homogeneous whole remains ω -negligible in size relative to the size of the whole. That is, the sum of such geographic-scale Q_Q -inhomogeneities has to be of a scale that is different from the scale of the whole.

From the constraints in Definition 3.1, it follows that the actual numerical differences in size are variable: the larger the ω - Q -homogeneous whole, the larger the Q_Q -inhomogeneities that are permitted. What exactly counts as a permitted Q_Q -inhomogeneity of a ω - Q -homogeneous geographic region depends on what counts as ω -roughly-the-same-size. This, in turn, depends on the precisification parameter $\omega \in \Omega$.

It follows from Lemma 3.2 and Definition 4.2 that if x is Q -homogeneous on a precisification ω_1 that allows for only small inhomogeneities, then x is also Q -homogeneous on a precisification ω_2 that allows for larger inhomogeneities:

Lemma 4.3: *If x is ω_1 - Q -homogeneous and $0 < \omega_1 \leq \omega_2 \leq 0.6$, then x is also ω_2 - Q -homogeneous.*

5. Quality pattern and regional partitions

In the next three sections the notion of homogeneity is used to develop the formal base of a mereology-based ontology of regional classification and delineation systems.

5.1. Relations between quality pattern

Definition 5.1: Let $Q_1 = \langle Q_1, \dots, Q_n \rangle$ and $Q_2 = \langle Q'_1, \dots, Q'_m \rangle$ be quality pattern. Q_1 is a sub-pattern of Q_2 ($Q_1 \sqsubseteq Q_2$) if and only if for all Q_k ($1 \leq k \leq n$) there is a Q'_j ($1 \leq j \leq m$) such that $Q_k = Q'_j$; Q_1 is a proper sub-pattern of Q_2 ($Q_1 \sqsubset Q_2$) if and only if $Q_1 \sqsubseteq Q_2$ and not $Q_2 \sqsubseteq Q_1$.

For example, $\langle Q_1 \rangle \sqsubset \langle Q_1, Q_2 \rangle$; $\langle Q_2, \dots, Q_{n-1} \rangle \sqsubset \langle Q_1, \dots, Q_n \rangle$; $\langle Q_1, \dots, Q_n \rangle \sqsubseteq \langle Q_1, \dots, Q_n \rangle$. Among the quality pattern displayed in Table 2, the sub-pattern relation holds as indicated in Figure 4.

Definition 5.2: Let $Q_S = \langle Q_1, \dots, Q_n \rangle$ and $Q_G = \langle Q'_1, \dots, Q'_n \rangle$ be quality pattern such that Q_j is a proper sub-universal-of Q'_j for $1 \leq j \leq n$, then Q_G is a genus pattern of Q_S and Q_S is a species pattern of Q_G .

Table 2. Genus quality pattern (based on Omernik 1987, (Table 1) and Bailey 1983).

Quality pattern	Symbol	Constituting quality universals
Land-surface	$Q-L$	$\langle \text{Land-surface form, Climax Plant Formation, Land use, Soil type} \rangle$
Climate	$Q-Do$	$\langle \text{Climate regime} \rangle$
	$Q-Di$	$\langle \text{Climate regime, Climate type, Elevation} \rangle$
	$Q-P$	$\langle \text{Climate regime, Climate type, Elevation, Climax Vegetation} \rangle$
Climate + Land-surface	$Q-S$	$\langle \text{Climate regime, Climate type, Elevation, Climax Vegetation, Land-surface form, Land use, Soil type} \rangle$

Table 3. Species quality pattern for some ecoregions of the North American Continent (based on Bailey 1983).

Quality pattern for the ecoregion	Symbol	Constituting quality universals	Genus pattern
Prairie division (of <i>NAC</i>)	<i>Q-Pra</i>	<i>⟨Humid Temperate Climate Regime, Prairie Climate Type, Lowland⟩</i>	<i>Q-Di</i>
Lowland tundra div. (of <i>NAC</i>)	<i>Q-TuL</i>	<i>⟨Polar Climate Regime, Tundra Climate Type, Lowland⟩</i>	<i>Q-Di</i>
Highland tundra div. (of <i>NAC</i>)	<i>Q-TuH</i>	<i>⟨Polar Climate Regime, Tundra Climate Type, Highland⟩</i>	<i>Q-Di</i>
Dry domain (of <i>NAC</i>)	<i>Q-Dry</i>	<i>⟨Dry Climate Regime⟩</i>	<i>Q-Do</i>
Humid temperate domain	<i>Q-HTe</i>	<i>⟨Humid Temperate Climate Regime⟩</i>	<i>Q-Do</i>
Humid tropical domain	<i>Q-HTr</i>	<i>⟨Humid Tropical Climate Regime⟩</i>	<i>Q-Do</i>
Polar domain	<i>Q-Pol</i>	<i>⟨Polar Climate Regime⟩</i>	<i>Q-Do</i>

In Table 1 the quality pattern in the body of the table are species quality pattern of the genus-pattern *Q-L*, which is defined in Table 2. The genus-pattern *Q-L* is also listed in the header of Table 1. The genus quality pattern that are used in the discussion of Bailey's system are listed in Table 2. Some of the relevant species quality pattern are listed in Table 3.

5.2. Quality pattern and partitions

Every homogeneity-based delineation system consists of a collection *p* of geographic regions, which forms a partition of (large i-parts of) the surface of the Earth. The members of *p* are all ω -homogeneous with respect to *distinct* quality pattern. Therefore their boundaries are located where there are inhomogeneities in the spatial distribution of quality pattern in the sense of Mark and Csillag (1989). If the quality patterns that characterize the members of *p* are all species pattern of a common genus pattern *Q*, then *p* is a ω -*Q*-partition of (i-parts of) the surface of the Earth. More precisely:

Definition 5.3: Collection *p* is a ω -*Q*-partition of the geographic region *x* if and only if (a) *p* is a partition of *x*, (b) every member of *p* is ω -*Q'*-homogeneous for some species quality pattern *Q'* of *Q*, and (c) if *y* and *z* are members of *p*, *y* is ω -*Q''*-homogeneous, *z* is ω -*Q''*-homogeneous, and *Q''* is a species quality pattern of *Q* then *y* = *z*.

If *p* is a *Q*-partition of *x* on a precisification ω_1 that allows for only small inhomogeneities, then *p* is also *Q*-partition of *x* on a precisification ω_2 that allows for larger inhomogeneities. That is, by Definition 5.3 and Lemma 4.3:

Lemma 5.4: If *p* is a ω_1 -*Q*-partition of *x* and $0 < \omega_1 \leq \omega_2 \leq 0.6$, then *p* is also ω_2 -*Q*-partition of *x*.

In a ω -*Q*-partition *p* of *x* every member of *p* is identified by a species pattern of *Q*. Because since in such a ω -*Q*-partition the boundaries of members of *p* coincide with discontinuities in the distribution of quality pattern, every member of *p* is maximally *Q'*-homogeneous with respect to some species pattern of *Q*. Suppose that there is a precisification parameter ω_{LIII} such that (i) the collection *LIII* forms a ω_{LIII} -*Q-L*-partition of the contiguous United States (*CUS*) and (ii) this partition roughly corresponds to the delineation depicted in the map in Figure 1c. On this assumption, every member of *LIII* is maximally- ω_{LIII} -homogeneous with respect to one quality pattern *Q'* as specified in Omernik (1987,

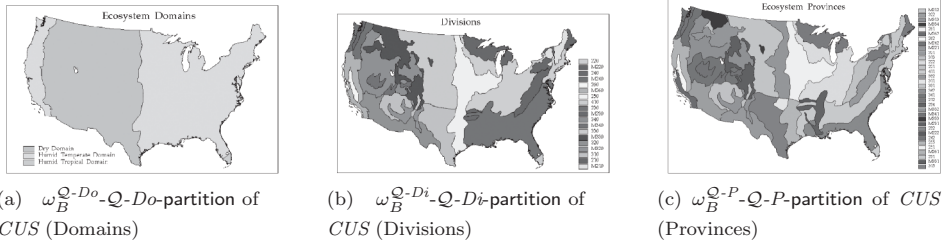


Figure 2. Ecoregions of the contiguous United States (CUS) at different scales USFS (1995).

Table 1): (i) Every two members of $LIII$ are ω_{LIII} -homogeneous with respect to distinct quality pattern and no two members of $LIII$ overlap; (ii) Jointly, the members of $LIII$ sum-up to CUS .

Consider Figure 2 and suppose that there are precisification parameters $\omega_B^{Q-D_o}$, $\omega_B^{Q-D_i}$, and ω_B^{Q-P} such that the regional delineations depicted in the maps respectively correspond to the following partitions of the contiguous USA (CUS): p_{do} , a $\omega_B^{Q-D_o}$ - Q - D_o -partition of CUS (Figure 2a); p_{di} , a $\omega_B^{Q-D_i}$ - Q - D_i -partition of CUS (Figure 2b), and p_{pr} , a ω_B^{Q-P} - Q - P -partition of CUS (Figure 2c). As pointed out in Section 4, the larger a Q -homogeneous region as a whole, the larger the permitted inhomogeneities. For reasons of simplicity, it therefore seems reasonable to stipulate that $\omega_B^{Q-D_o} = \omega_B^{Q-D_i} = \omega_B^{Q-P} = \omega_B$.

In Figure 2a one consequence of the maximal- $\omega_B^{Q-D_o}$ - Q - D_o' -homogeneity of each member of the depicted $\omega_B^{Q-D_o}$ - Q - D_o -partition ($Q-D_o' \sqsubset Q-D_o$) is that there may be members that consist of multiple topologically disconnected i-parts. For example, p_{do} has three members, one of which has two disconnected parts.

Consider the partitions formed by the collections p_{do} , p_{di} , and p_{pr} as depicted in Figure 2. Table 2 shows that the following sub-pattern relations hold between the quality pattern of the underlying partitions: $Q-D_o \sqsubset Q-D_i \sqsubset Q-P$. Correspondingly, the respective partitions are nested hierarchically, that is, p_{pr} is $\uparrow p$ -included-in p_{di} and p_{di} is $\uparrow p$ -included-in p_{do} . In general, it seems that super-patterns of a given quality pattern often generate finer partitions than the given sub-pattern.

Definition 5.5: Let p_1 be a ω_1 - Q_1 -partition of x and let p_2 be a ω_2 - Q_2 -partition of x . Partition p_1 is refined by p_2 if and only if $Q_1 \sqsubset Q_2$, p_2 $\uparrow p$ -included-in p_1 , and not p_1 $\uparrow p$ -included-in p_2 .

For example, partition p_{do} is refined by p_{di} , which is refined by p_{pr} .

The following assumption is made to simplify the presentation of the running example:

Assumption 5.6 Let SEC be the collection of subregional ecoregions (sections) identified by Bailey (1983) and more precisely characterized by the USDA Forest Service (1992). SEC as well as the collection $LIII$ of the EPA (CEC 1997) form partitions whose members are maximally homogeneous with respect to the quality pattern listed in Omernik (1987; Table 1).

Indeed, one can verify that most of the qualities listed by Omernik (1987; Table 1) are also identified by the USDA Forest Service (1992). In conjunction with Definition 5.3, Assumption 5.6 entails that the quality pattern identified by Omernik (1987; Table 1) are sufficient (for the purpose of the examples) to *identify/distinguish* the right ecoregions in both systems (Bailey/USFS and EPA). For example, both systems identify/distinguish regions such as Central Great Plains (CGP), and Flint Hills (FH). Assumption 5.6 does *not* entail that these quality patterns are sufficient to delineate such regions precisely. This

will be discussed in detail in Section 6. That Assumption 5.6 is a simplification can be seen, for example, from the fact that the ecoregion High Plains (*HP*) of (Omernik 1987), a member of *LIII*, roughly corresponds to the sum of the ecoregions Southern High Plains, Central Table Lands, and Central High Plains of the USDA Forest Service (1992) (members of *SEC*).

5.3. Local and nonlocal quality pattern

Using the notions of homogeneity, partition, and relations between quality pattern, local and nonlocal quality pattern can be distinguished.

Definition 5.7: (i) A quality pattern Q is nonlocal in the partition of the geographic region x formed by the collection p on the precisification ω (Q is ω -nonlocal in p) if and only if p is a ω - Q -partition of x . (ii) Q is ω -global in p if and only if Q is ω -nonlocal in p and x is a region of continental scale or larger.

For example, the quality pattern Q - L is global in the partition of CUS formed by the collection *LIII* on the precisification ω_{LIII} , because *LIII* is a ω_{LIII} - Q - L -partition of CUS and CUS is a region of continental scale. Similarly, in Table 2, as all the quality pattern in the table generate partitions of CUS (displayed in Figures 1c and 2), it follows that all quality pattern are ω_B -global in the respective partitions of CUS .

Definition 5.8: (i) A quality pattern Q is local in the partition of x formed by the collection p on the precisification ω (Q is ω -local in p) if and only if (a) p is a partition of x but not a ω - Q -partition of x and (b) there is a sub-collection q of p such that q is a ω - Q -partition of y . (ii) Q is ω -regional in p if and only if Q is ω -local in p and y is of regional scale or smaller and x is of continental scale or larger.

Consider Table 4 that corresponds to Table 1 but adds additional qualities as detailed in EPA (2002). Table 4 shows that the additional qualities of a given ecoregion are rather specific and are often specified relative to the qualities of its neighbors (e.g., lower/higher precipitation, less relief, more irregular). In addition, historical qualities of certain ecoregions are included (e.g., ‘Once a grassland ...’). Consider the ecoregion Central Great Plains. From the qualities listed in Tables 1 and 4 one can conclude that this region is maximally homogeneous with respect to a species pattern of the quality pattern:

$$Q_{WHP} = \langle LF, CV, LU, S, \text{Relief-difference-to-Western-High-Plains}, \text{Precipitation-difference-to-Western-High-Plains}, \text{Past-land-use} \rangle. \quad (1)$$

Table 4. Additional local qualities of two members of the collection *LIII* EPA (2002).

Geographic region	Quality pattern
...	...
Central great plains	... are slightly lower, receive more precipitation, and are somewhat more irregular than the Western High Plains to the west ... Once a grassland, with scattered low trees and shrubs in the south, much of this ecological region is now cropland ...
Flint Hills	rolling hills with relatively narrow steep valleys ... composed of shale and cherty limestone with rocky soils ... mark the western edge of the tallgrass prairie ... In contrast to surrounding ecological regions that are mostly in cropland, most of the Flint Hills region is grazed by beef cattle ...
...	...

The quality pattern Q_{WHP} is a super-pattern of $Q-L$ and includes additional quality universals. Q_{WHP} is regional (and thus local) in the partition of CUS formed by $LIII$ on the precisification ω_{LIII} : there is a sub-collection q of $LIII$ such that q is a ω_{LIII} - Q_{WHP} -partition of some i -part of CUS which is of regional scale or smaller. In this specific case q is probably identical to the collection that has Central Great Plains as its only member, that is, $q = \{\text{Central Great Plains}\}$.

Note that there is also a collection r such that r is a ω_{LIII} - Q_{WHP} -partition of CUS . However, the partition formed by r is *distinct* from the partition formed by $LIII$: Q_{WHP} is a super-pattern of $Q-L$ and the resulting partition refines the partition formed by $LIII$ in the sense of Definition 5.5. This example shows that there are quality patterns that are ω -local in one partition of a region x and ω -nonlocal in another partition of x .

6. Vague partitions

Consider Figure 3 in which two ways of crisply delineating the CUS into maximally homogeneous subregions – ecoregions of subregional scale – are (partly) displayed. The bold black lines depict the delineation formed by the collection SEC , the ecoregion sections of Bailey (1983). The nonbold black lines depict the delineation formed by the collection $LIII$ of the CEC (1997). Assume that SEC is a ω_B - $Q-L$ -partition of CUS and that $LIII$ is a ω_{LIII} - $Q-L$ -partition of CUS . On the semantic view of vagueness one can understand SEC and $LIII$ as two precisifications of one vague Ω - $Q-L$ -partition of CUS with $\omega_B, \omega_{LIII} \in \Omega$. That is, the two partitions of Figure 3 represent different ways of making the vague, homogeneity-based delineation of ecoregions of subregional scale precise. This point will be developed more precisely in this section.

6.1. Precisification ranges and vague partitions

At the formal level the notation of ‘ Ω - Q -partition of x ’ is introduced to refer to the vague partition of x , which is characterized by the precisification range Ω for the precisification parameter ω and the genus quality pattern Q . The following axiom is included:

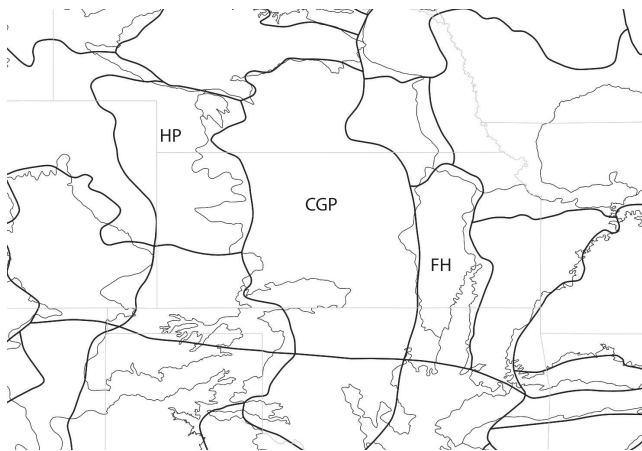


Figure 3. Bold black boundaries: the ω_B - $Q-L$ -partition of CUS (ecoregion sections) of Bailey (1983); Nonbold black boundaries: the ω_{LIII} - $Q-L$ -partition of CUS (members of $LIII$) of the CEC (1997). (The maps are taken from the National Atlas (CEC 2009).)

Axiom 6.1 If a vague Ω - $\mathcal{Q}$ -partition of x is such that $\Omega = [\omega_{\downarrow}, \omega_{\uparrow}]$, then for all ω such that $\omega_{\downarrow} \leq \omega \leq \omega_{\uparrow}$, there exists (at least) one collection p_{ω} such that p_{ω} is a ω - $\mathcal{Q}$ -partition of x .

Semantically, a vague Ω - $\mathcal{Q}$ -partition of x is interpreted as the set of all the collections that form crisp ω - $\mathcal{Q}$ -partitions of x such that $\omega \in \Omega$. Let $\mathcal{P}_{(\mathcal{Q}, \omega)}^x$ be the set of precisifications of the vague Ω - $\mathcal{Q}$ -partition of x , that is,

$$\mathcal{P}_{(\mathcal{Q}, \Omega)}^x =_{df} \{p | p \text{ is a } \omega\text{-}\mathcal{Q}\text{-partition of } x \text{ for some } \omega \text{ such that } \omega_{\downarrow} \leq \omega \leq \omega_{\uparrow}, \text{ and } \Omega = [\omega_{\downarrow}, \omega_{\uparrow}]\}. \quad (2)$$

On this interpretation, the parameters Ω and $\mathcal{Q}$ uniquely identify one particular vague partition of x and the corresponding set $\mathcal{P}_{(\mathcal{Q}, \Omega)}^x$. That is, the notation ‘the Ω - $\mathcal{Q}$ -partition of x ’ can be used to refer to the vague partition of x that is characterized by Ω and $\mathcal{Q}$. The precisification parameter $\omega_{\downarrow}$ determines the largest lower bound of the degrees of inhomogeneity permitted in the underlying vague Ω - $\mathcal{Q}$ -partition. The precisification parameter $\omega_{\uparrow}$ determines the least upper bound of the degrees of inhomogeneity permitted in the underlying vague $\mathcal{Q}$ -partition.

This interpretation reflects two important aspects of the semantic understanding of the granularity- and-homogeneity-based vagueness of a Ω - $\mathcal{Q}$ -partition of some region x : (i) The different ways, $\omega \in \Omega$, of making the meaning of the vague predicate $\mathcal{Q}'$ -homogeneous precise for the various species pattern $\mathcal{Q}'$ of $\mathcal{Q}$ are all semantically equivalent; correspondingly, (ii) the geometrically and topologically different ways of delineating a certain class of maximally homogeneous subregions of x , that is, the collections forming (crisp) ω - $\mathcal{Q}$ -partitions of x in $\mathcal{P}_{(\mathcal{Q}, \Omega)}^x$, are all semantically equivalent. Consider, for example, the vague Ω - $\mathcal{Q}$ - L -partition of *CUS*. If $\omega_B, \omega_{LIII} \in \Omega$, then the collections *SEC* and *LIII*, depicted in Figure 3, are both equivalent with respect to the semantic understanding of the granularity- and- homogeneity-based vagueness of the vague Ω - $\mathcal{Q}$ - L -partition of *CUS*. That is, $SEC, LIII \in \mathcal{P}_{(\mathcal{Q}-L, \Omega)}^{CUS}$.

In Figure 3, note that the claim here is not that variations of the precisification parameter ω alone can account for all of the topological and geometrical differences between the two delineations depicted in the figure. The focus here is on *granularity-and-homogeneity-based* aspects of vagueness. The claim is that (i) the *granularity-and-homogeneity-based* aspects of vagueness are captured (to a reasonable degree) in the precisification parameters and ranges and the interpretation of the corresponding vague partitions, and (ii) the captured granularity-homogeneity-based aspects of vagueness contribute to a significant degree to the differences in the depicted delineations. Of course, there are also other contributing factors including arbitrary human map generalization and differences in the techniques that were used to produce the corresponding maps, etc.

6.2. Representing vague partitions

From the perspective of the representation of vague partitions, the semantic view of vagueness – according to which all admissible precisification parameters $\omega \in \Omega$ and, correspondingly, all admissible delineations $p \in \mathcal{P}_{(\mathcal{Q}, \Omega)}^x$ of a vague Ω - $\mathcal{Q}$ -partition are semantically equivalent – seems to conflict with gradation-based views (e.g., Kronenfeld 2005). According to gradation-based views, some delineations of a vague Ω - $\mathcal{Q}$ -partition are better suited for representation purposes than others. The preference of some delineations and, correspondingly, some precisification parameters over others gives rise to a gradation. It then

seems to follow that, according to this view, not all precisification parameters $\omega \in \Omega$ and not all delineations $p \in \mathcal{P}_{(\mathcal{Q}, \Omega)}^x$ are semantically equivalent.

In addition, an inspection of Figure 3 immediately reveals that *visually* there is an obvious difference between the generalized and coarse character of Bailey's delineation (bold black boundaries) and the more fine-grainedness of the EPA delineation (nonbold black boundaries). Visually, the two delineations do *not* seem to be semantically equivalent precisifications of a single vague partition – the vague Ω - $\mathcal{Q}$ - L -partition of *CUS* – but each delineation seems to be interpreted as a precisification that serves as a geometric representation for one of *two* vague partitions. More precisely:

Proposition 6.2: (1) *The delineation formed by the collection SEC, a ω_B - $\mathcal{Q}$ - L -partition of CUS, represents the vague Ω_B - $\mathcal{Q}$ - L -partition of CUS with $\Omega_B = [\omega_{\downarrow}^B, \omega_B]$; (2) *The delineation formed by the collection LIII, a ω_{LIII} - $\mathcal{Q}$ - L -partition of CUS, represents the vague Ω_O - $\mathcal{Q}$ - L -partition of CUS with $\Omega_O = [\omega_{\downarrow}^O, \omega_{LIII}]$; (3) *The inhomogeneities permitted in the vague Ω_B - $\mathcal{Q}$ - L -partition of CUS are larger than the inhomogeneities permitted in the Ω_O - $\mathcal{Q}$ - L -partition of CUS. The inhomogeneities permitted in the Ω_B - $\mathcal{Q}$ - L -partition are smaller or equal to the inhomogeneities permitted in the Ω - $\mathcal{Q}$ - L -partition of CUS.***

The following argument can be given to support this claim from a cartographic/representational perspective. The visual cue for the larger degree of inhomogeneity permitted by the Ω_B - $\mathcal{Q}$ - L -partition of *CUS* seems to be the more smooth and generalized character of the boundaries of the bold black delineation in Figure 3. That is, smooth and highly generalized boundaries seem to convey the intuition that there is a large degree of ‘freedom’ to place boundaries by fiat in the way that supports best the purpose of a map as a medium for conveying information. As Kronenfeld (2005) puts it, to minimize information decoding error, map designers strive (i) for crisp (nongraduated) boundaries and (ii) for minimizing boundary complexity by drawing boundaries in highly generalized ways. From the perspective of representing vague partitions, drawing boundaries in highly generalized ways avoids the misinterpretation of vague boundaries that are crisped by fiat as realism of a map.

Proposition 6.2 is a specific instance of the more general claim that admissible representations for a vague Ω - $\mathcal{Q}$ -partition of x are collections that form a $\mathcal{Q}$ -partition of x that is consistent with the *largest* inhomogeneities permitted by Ω :

Axiom 6.3 Collection p is an admissible representation for the vague $[\omega_{\downarrow}, \omega_{\uparrow}]$ - $\mathcal{Q}$ -partition of x if and only if p is a $\omega_{\uparrow}$ - $\mathcal{Q}$ -partition of x .

Semantically, Axiom 6.3 can be motivated as follows: For any choice of $\omega \in \Omega$ there is a set of collections that form a ω - $\mathcal{Q}$ -partition of x such that the generated delineation of subregions of x is consistent with the inhomogeneities permitted by ω . Let $\mathcal{P}_{(\mathcal{Q}, \omega)}^x$ be this set, that is,

$$\mathcal{P}_{(\mathcal{Q}, \omega)}^x =_{df} \{p \mid p \text{ is a } \omega\text{-}\mathcal{Q}\text{-partition of } x\}. \quad (3)$$

Consider the sets $\mathcal{P}_{(\mathcal{Q}, \omega_1)}^x$ and $\mathcal{P}_{(\mathcal{Q}, \omega_2)}^x$. The larger the precisification parameter ω , the larger the inhomogeneities that are tolerated by the underlying notion of ω - $\mathcal{Q}$ -homogeneity. Consequently, the larger the degree of freedom to place boundaries and the larger the degree of vagueness of boundary location, and the larger the set of ω - $\mathcal{Q}$ -partitions ($\mathcal{P}_{(\mathcal{Q}, \omega)}^x$). More precisely:

Lemma 6.4: *If $0 < \omega_1 \leq \omega_2 \leq 0.6$ then $\mathcal{P}_{(\mathcal{Q}, \omega_1)}^x \subseteq \mathcal{P}_{(\mathcal{Q}, \omega_2)}^x$.*

Proof: Assume that $0 < \omega_1 \leq \omega_2 \leq 0.6$. One has to show that if $p \in \mathcal{P}_{(\mathcal{Q}, \omega_1)}^x$, then $p \in \mathcal{P}_{(\mathcal{Q}, \omega_2)}^x$. Assume that $p \in \mathcal{P}_{(\mathcal{Q}, \omega_1)}^x$. Thus p is a ω_1 - $\mathcal{Q}$ -partition of x . By Lemma 5.4, p is also a ω_2 - $\mathcal{Q}$ -partition of x . Hence, $p \in \mathcal{P}_{(\mathcal{Q}, \omega_2)}^x$. $\square$

Now consider the vague Ω - $\mathcal{Q}$ -partition of x with $\Omega = [\omega_{\downarrow}, \omega_{\uparrow}]$. As specified in Equation (2), this vague partition can be semantically identified with the set of all collections that form ω - $\mathcal{Q}$ -partitions of x such that $\omega_{\downarrow} \leq \omega \leq \omega_{\uparrow}$. This is equivalent to

$$\mathcal{P}_{(\mathcal{Q}, \Omega)}^x = \bigcup_{\omega \in \Omega} \mathcal{P}_{(\mathcal{Q}, \omega)}^x. \quad (4)$$

By Lemma 6.4, $\mathcal{P}_{(\mathcal{Q}, \omega)}^x \subseteq \mathcal{P}_{(\mathcal{Q}, \omega_1)}^x$, for all $\omega \in \Omega$. Thus, $\mathcal{P}_{(\mathcal{Q}, \Omega)}^x \subseteq \mathcal{P}_{(\mathcal{Q}, \omega_1)}^x$. Hence, $\mathcal{P}_{(\mathcal{Q}, \Omega)}^x = \mathcal{P}_{(\mathcal{Q}, \omega_1)}^x$. The identity of these two sets supports the claim of Axiom 6.3 that every collection p that is a $\omega_{\uparrow}$ - $\mathcal{Q}$ -partition of x is an admissible representation for the vague Ω - $\mathcal{Q}$ -partition of x . From the fact that p is a $\omega_{\uparrow}$ - $\mathcal{Q}$ -partition of x one can ‘reconstruct’ the set of all admissible precisifications of the vague Ω - $\mathcal{Q}$ -partition of x . This analysis also shows that, for representation purposes and the analysis of the vagueness of boundary locations, $\omega_{\downarrow}$, the greatest lower bound of the permitted inhomogeneity can be ignored.

6.3. Relations between vague partitions

The binary predicates compatible, less-or-equally-inhomogeneous, less-or-equally-vague, and refined-by are introduced to compare and relate various aspects of vague partitions. Let Ω_1 and Ω_2 be the precisification ranges of two vague partitions. Then $\Omega_1 \preceq \Omega_2$ if and only if $\Omega_1 = [\omega_{\downarrow}^1, \omega_{\uparrow}^1]$ and $\Omega_2 = [\omega_{\downarrow}^2, \omega_{\uparrow}^2]$ and $\omega_{\downarrow}^1 \leq \omega_{\downarrow}^2$.

Definition 6.5: (i) The Ω_1 - $\mathcal{Q}_1$ -partition of x is compatible with the Ω_2 - $\mathcal{Q}_2$ -partition of y if and only if $x = y$ and either $\mathcal{Q}_1 \sqsubseteq \mathcal{Q}_2$ or $\mathcal{Q}_2 \sqsubseteq \mathcal{Q}_1$. (ii) The Ω_1 - $\mathcal{Q}_1$ -partition of x is less-or-equally-inhomogeneous than the Ω_2 - $\mathcal{Q}_2$ -partition of x if and only if $\mathcal{Q}_1 \sqsubseteq \mathcal{Q}_2$ and $\Omega_1 \preceq \Omega_2$. (iii) The Ω_1 - $\mathcal{Q}_1$ -partition of x is less-or-equally-vague than the Ω_2 - $\mathcal{Q}_2$ -partition of x if and only if $\mathcal{P}_{(\mathcal{Q}_1, \Omega_1)}^x \subseteq \mathcal{P}_{(\mathcal{Q}_2, \Omega_2)}^x$. (iv) The Ω_1 - $\mathcal{Q}_1$ -partition of x is refined-by the Ω_2 - $\mathcal{Q}_2$ -partition of x if and only if $\Omega_1 = [\omega_{\downarrow}^1, \omega_{\uparrow}^1]$ and $\Omega_2 = [\omega_{\downarrow}^2, \omega_{\uparrow}^2]$ and there are collections p_1 and p_2 such that p_1 is a $\omega_{\downarrow}^1$ - $\mathcal{Q}_1$ -partition of x and p_2 is a $\omega_{\downarrow}^2$ - $\mathcal{Q}_2$ -partition of x and p_1 is refined-by p_2 in the sense of Definition 5.5.

Compatible vague partitions subdivide identical geographic regions and their quality pattern are sub-pattern/super-pattern of one another. For example, the vague partitions depicted in Figures 1c and 2 are all compatible: They partition CUS and $\mathcal{Q}\text{-}Do \sqsubset \mathcal{Q}\text{-}Di \sqsubset \mathcal{Q}\text{-}P \sqsubset \mathcal{Q}\text{-}L$. The discussion below will mostly focus on compatible vague partitions.

One can prove that for compatible vague partitions with identical quality pattern, less inhomogeneity implies less vagueness:

Lemma 6.6: *If the Ω_1 - $\mathcal{Q}$ -partition of x is less-or-equally-inhomogeneous than the Ω_2 - $\mathcal{Q}$ -partition of x , then the Ω_1 - $\mathcal{Q}$ -partition of x is less-or-equally-vague than the Ω_2 - $\mathcal{Q}$ -partition of x .*

Proof: Assume that the Ω_1 - $\mathcal{Q}$ -partition of x with $\Omega_1 = [\omega_{\downarrow}^1, \omega_{\uparrow}^1]$ is less-or-equally-inhomogeneous than the vague Ω_2 - $\mathcal{Q}$ -partition of x with $\Omega_2 = [\omega_{\downarrow}^2, \omega_{\uparrow}^2]$. That is, $\omega_{\downarrow}^1 \leq \omega_{\downarrow}^2$. One has to show: if $p \in \mathcal{P}_{(\mathcal{Q}, \Omega_1)}^x$ then $p \in \mathcal{P}_{(\mathcal{Q}, \Omega_2)}^x$. Assume $p \in \mathcal{P}_{(\mathcal{Q}, \Omega_1)}^x$. Then p is a ω - $\mathcal{Q}$ -partition of x such that $\omega_{\downarrow}^1 \leq \omega \leq \omega_{\uparrow}^1$. Since $\omega_{\downarrow}^1 \leq \omega_{\downarrow}^2$, $\omega \leq \omega_{\uparrow}^2$ and, by Lemma 5.4, p is a $\omega_{\downarrow}^2$ - $\mathcal{Q}$ -partition of x . That is, $p \in \mathcal{P}_{(\mathcal{Q}, \Omega_2)}^x$. $\square$

For example, from Proposition 6.2 it follows that the vague Ω_O - Q - L -partition of CUS represented by the collection $LIII$ is less-or-equally-inhomogeneous and less-or-equally-vague compared with the vague Ω_B - Q - L -partition of CUS represented by the collection SEC .

Considering Figure 2, the collection p_{do} depicted in Figure 2a is a ω_B - Q - Do -partition of CUS and represents the vague $[\omega_{\downarrow}^{Q-Do}, \omega_B]$ - Q - Do -partition of CUS ; the collection p_{di} (Figure 2b) is a ω_B - Q - Di -partition of CUS and represents the vague $[\omega_{\downarrow}^{Q-Di}, \omega_B]$ - Q - Di -partition of CUS ; the collection p_{pr} (Figure 2c) is a ω_B - Q - P -partition of CUS and represents the vague $[\omega_{\downarrow}^{Q-P}, \omega_B]$ - Q - P -partition of CUS . The $[\omega_{\downarrow}^{Q-Do}, \omega_B]$ - Q - Do -partition of CUS is refined-by the $[\omega_{\downarrow}^{Q-Di}, \omega_B]$ - Q - Di -partition of CUS , which in turn is refined-by the $[\omega_{\downarrow}^{Q-P}, \omega_B]$ - Q - P -partition of CUS . This is because the respective representing crisp partitions p_{do} , p_{di} , and p_{pr} refine one another as pointed out in the end of Section 5.2.

7. Quality pattern, inhomogeneity, and vagueness

Delineation systems for geographic regions usually aim for delineating regions as precisely as possible. That is, they aim for identifying vague partitions that are minimally inhomogeneous and minimally vague. For this purpose, it is important to understand the logical interrelations between local and nonlocal quality pattern, inhomogeneity, and vagueness in compatible vague partitions. Let the Ω_1 - Q_1 -partition of x and the Ω_2 - Q_2 -partition of x be two compatible vague partitions. The aim of this section is to show that, from an ontological perspective, the Ω_1 - Q_1 -partition of x is less-or-equally-inhomogeneous than the Ω_2 - Q_2 -partition of x only if one of two conditions are met. First, Q_1 is a nonlocal super-pattern of Q_2 . That is, additional nonlocal qualities are responsible for the lower degree of inhomogeneities in the first partition. Second, $Q_1 = Q_2$ and there are additional local super pattern $Q'_1, \dots, Q'_n$ of Q_1 and these additional local qualities are responsible for the lower degree of inhomogeneities permitted by the Ω_1 - Q_1 -partition of x . From Lemma 6.6 it then follows that Ω_1 - Q_1 -partition of x is also less-or-equally-vague compared to the Ω_2 - Q_2 -partition of x . Both cases are discussed in sequence.

7.1. Inhomogeneity and nonlocal quality pattern

It seems reasonable to assume that adding additional nonlocal qualities to the quality pattern of a vague partition of x gives rise to a less-or-equally-inhomogeneous vague partition of x . For example, the vague Ω_{Q-S} - Q - S -partition of CUS , which is formed by regions that are maximally homogeneous with respect to climate qualities *and* land-surface form qualities, will be less-or-equally-inhomogeneous compared to the vague Ω_{Q-P} - Q - P -partition of CUS , which is formed by regions that are maximally homogeneous only with respect to climate qualities. Formally, this is expressed in Axiom 7.1:

Axiom 7.1 If the Ω_1 - Q_1 -partition of x and the Ω_2 - Q_2 -partition of x are such that $Q_1 \sqsubseteq Q_2$, then the Ω_2 - Q_2 -partition of x is less-or-equally-inhomogeneous compared to the Ω_1 - Q_1 -partition of x (i.e., $\Omega_2 \leq \Omega_1$).

Consider the vague partitions depicted in Figure 2. Because $Q-Do \sqsubset Q-Di \sqsubset Q-P$, Axiom 7.1 permits to infer that $\Omega_{Q-P} \preceq \Omega_{Q-Di} \preceq \Omega_{Q-Do}$.

Consider the Ω_1 - Q_1 -partition of x and the Ω_2 - Q_2 -partition of x . If $Q_2 \sqsubset Q_1$ then the Ω_1 - Q_1 -partition of x will often be a refinement of the Ω_2 - Q_2 -partition of x in the sense of Definition 6.5(iv). That is, nonlocal super-pattern such as Q_1 usually characterize Ω_1 - Q_1 -partitions of x that are *more fine grained* and *less inhomogeneous* than partitions that are based on sub-pattern of Q_1 such as the Ω_2 - Q_2 -partition of x . For example, the Ω_{Q-P} - Q - P -partition of CUS refines the Ω_{Q-Di} - Q - Di -partition of CUS and is, in addition, less-or-equally-

inhomogeneous compared to this partition. The Ω_{Q-Di} - Q - Di -partition of CUS , in turn, refines and is less-or-equally-inhomogeneous compared to the Ω_{Q-Do} - Q - Do -partition of CUS , etc.

7.2. Inhomogeneity and local quality pattern

Consider the vague Ω_B - Q - L -partition of CUS with $\Omega_B = [\omega_{\downarrow}^B, \omega_B]$. For every $\omega \in \Omega_B$, the members of the ω - Q - L -partition of CUS are ω -homogeneous with respect to quality pattern as specified in Omernik (1987; Table 1). In conjunction with Proposition 6.2, it seems reasonable to assume that $\omega_{\downarrow}^B \leq \omega_{LIII} < \omega_B$. That is, SEC , the ω_B - Q - L -partition of CUS , as well as $LIII$, the ω_{LIII} - Q - L -partition of CUS , are precisifications of the vague Ω_B - Q - L -partition of CUS , that is, $SEC, LIII \in \mathcal{P}_{Q-L, \Omega_B}^{CUS}$.

Now consider the Ω_O - Q - L -partition of CUS with $\Omega_O = [\omega_{\downarrow}^O, \omega_{LIII}]$. For every $\omega \in \Omega_O$ the members of the respective precisification are ω -homogeneous with respect to the quality pattern specified in Omernik (1987; Table 1) and the quality pattern specified in EPA (2002). As discussed in Section 5.3, compared to the non-local quality pattern specified in Omernik (1987; Table 1), the quality pattern specified in EPA (2002) include additional local qualities. These local qualities reduce the inhomogeneities permitted in the Ω_O - Q - L -partition of CUS and, correspondingly, the degree of its vagueness such that $\mathcal{P}_{Q-L, \Omega_O}^{CUS} \subset \mathcal{P}_{Q-L, \Omega_B}^{CUS}$. Consider, for example, the bold black and nonbold black boundaries surrounding the label CGP in Figure 3. The region with nonbold black boundaries is ω_{LIII} - $QWHP$ -homogeneous and observations ‘on the ground’ have verified the consistency of this boundary with local variations of qualities such as precipitation and relief that separate the ecoregions Central Great Plains and High Plains (EPA 2002). That is, the particular observations of Central Great Plains support that $LIII \in \mathcal{P}_{Q-L, \Omega_B}^{CUS}$. By contrast, the inhomogeneities tolerated on the precisification parameter ω_B admit a delineation marked by the bold black boundaries of Figure 3. In particular toward the West, this specific boundary may not be consistent with the particular variation of the local qualities of precipitation and relief that establish the boundary between Central Great Plains and High Plains ‘on the ground’. If this is the case, then $SEC \notin \mathcal{P}_{Q-L, \Omega_O}^{CUS}$.

Given these intuitions, it seems reasonable to say that the vague Ω_O - Q - L -partition of CUS – in conjunction with the local quality pattern $\mathcal{Q}_1 \dots \mathcal{Q}_n$ specified in EPA (2002) – is a *locally crisped* version of the vague Ω_B - Q - L -partition of CUS (with the nonlocal quality pattern as specified in Omernik (1987; Table 1)). The notion of a locally crisped version of a vague partition is defined:

Definition 7.2: The vague Ω' - Q -partition of x is a locally crisped version of the vague Ω - Q -partition of x if and only if (a) $\Omega' \prec \Omega$; (b) there are quality pattern $\mathcal{Q}_1 \dots \mathcal{Q}_n$ such that for $1 \leq i \leq n$: (i) $\mathcal{Q} \sqsubset \mathcal{Q}_i$, (ii) for all $\omega \in \Omega'$ if p is a ω - Q partition of x then $\mathcal{Q}_i$ is ω -local in p , (iii) for all ω such that $\omega_{\downarrow}^{\downarrow} < \omega \leq \omega_{\uparrow}$ if p is a ω - Q partition of x then $\mathcal{Q}_i$ is neither ω -nonlocal in p nor ω -local in p .

To formally link the notions of locally crisped partition and less-inhomogeneous, the following axiom is included:

Axiom 7.3 The Ω_1 - Q -partition of x is less-inhomogeneous than the Ω_2 - Q -partition of x (i.e., $\Omega_1 \prec \Omega_2$), if and only if the Ω_1 - Q -partition of x is a locally crisped version of the Ω_2 - Q -partition of x .

That is, compatible vague partitions with identical quality pattern are equally inhomogeneous and equally vague *unless* other local qualities differentiate them such that one is a locally crisped version of the other.

It immediately follows from Lemma 6.6 that if the Ω_1 - $\mathcal{Q}$ -partition of x is a locally crisped version of the Ω_2 - $\mathcal{Q}$ -partition of x then the former is less vague than the latter, that is, $\mathcal{P}_{(\mathcal{Q}, \Omega_1)}^x \subset \mathcal{P}_{(\mathcal{Q}, \Omega_2)}^x$. From Axiom 7.3 and Lemma 6.6, it also follows:

Lemma 7.4: *If the vague Ω_1 - $\mathcal{Q}$ -partition of x is not a locally crisped version of the vague Ω_2 - $\mathcal{Q}$ -partition of x , then the Ω_2 - $\mathcal{Q}$ -partition of x is less-or-equally-inhomogeneous and less-or-equally-vague compared to the Ω_1 - $\mathcal{Q}$ -partition of x . I.e., $\Omega_2 \preceq \Omega_1$ and $\mathcal{P}_{(\mathcal{Q}, \Omega_2)}^x \subseteq \mathcal{P}_{(\mathcal{Q}, \Omega_1)}^x$.*

That is, vague Ω - $\mathcal{Q}$ -partitions that are *not* locally crisped are *maximally inhomogeneous* with respect to the quality pattern $\mathcal{Q}$ and *maximally vague* in the sense that they incorporate all semantically admissible ways of precisifying boundaries that are consistent with those inhomogeneities.

8. The classification of geographic regions based on homogeneity and nonlocal quality pattern

The focus of the discussion above was on the delineation of geographic regions. In this section the classification of geographic regions is discussed. Regional classification systems describe how sub-universal relations structure the universal *Georegion*. Sub-universals of *Georegion* include *Ecoregion*, *Biome*, *Ecozone*, and so on. In this section, *Georegion* universals are introduced formally. Then the Aristotelian method of classification is used to identify the sub-universals of *Georegion*. Finally, Bailey's ecoregion classification and delineation system is discussed as an example.

8.1. Georegions

For the discussion of the delineation of geographic regions it was sufficient to use the notion of *Georegion* somewhat informally. Before one can address the subject of classification, one needs to specify more precisely what is meant by *Georegion*. Intuitively, '*Georegion*' is a term referring to a universal that instances are all homogeneous with respect to some global quality pattern or a species pattern of a global quality pattern.

Because of the vague nature of the predicate homogeneous, it is necessary to refer to *Georegion* universals in precisification-dependent ways. For this purpose the notion ω - $\mathcal{Q}$ -*Georegion* universal is introduced. To formally link the notions of ω - $\mathcal{Q}$ -*Georegion* universal and ω - $\mathcal{Q}$ -homogeneity, the following axiom is included:

Axiom 8.1 If a is the universal referred to by the term ω - $\mathcal{Q}$ -*Georegion*, then (i) all instances of a are ω - $\mathcal{Q}$ -homogeneous, and (ii) there exists a ω - $\mathcal{Q}'$ -partition p of some i -part of the surface of the Earth, such that $\mathcal{Q}'$ is ω -global in p and either $\mathcal{Q} = \mathcal{Q}'$ or $\mathcal{Q}$ is a species pattern of $\mathcal{Q}'$.

Consequently, not every ω - $\mathcal{Q}$ -homogeneous geographic region is an instance of some ω - $\mathcal{Q}$ -*Georegion* universal: the quality pattern $\mathcal{Q}$ has to be globally relevant, that is, there has to be a partition p such that $\mathcal{Q}$ or genus pattern of $\mathcal{Q}$ is ω -global in p .

Particular x may be an instance of the universal referred to by the term $\mathcal{Q}$ -*Georegion* on the precisification ω_1 and not be an instance of the universal referred to by the term $\mathcal{Q}$ -*Georegion* on another precisification ω_2 . For example, in Figure 3 the bold black lines surrounding the label *CGP* represent the location of the boundary of the georegion CGP_{bd} . This region is ω_B - $\mathcal{Q}$ - $\mathcal{L}'$ -homogeneous, where $\mathcal{Q}$ - $\mathcal{L}'$ is the species pattern of $\mathcal{Q}$ - $\mathcal{L}$ identified in Table 1. CGP_{bd} is an instance of the universal referred to by the term ' ω_B - $\mathcal{Q}$ - $\mathcal{L}'$ -*Georegion*.' By contrast, CGP_{bd} is not an instance of the universal referred to by the term ' ω_{LIII} - $\mathcal{Q}$ - $\mathcal{L}'$ -*Georegion*.' The

instances of this universal are ω_{LIII} - Q - L' -homogeneous and therefore characterized by smaller inhomogeneities. Because the universals referred to by the terms ω_{LIII} - Q - L' -*Georegion* and ω_B - Q - L' -*Georegion* have different instances, they cannot be identical.

It follows that the term Q_1 -*Georegion* may refer to a sub-universal-of the universal referred to by the term Q_2 -*Georegion* on some precisifications but not on others. Because the underlying vagueness is captured by the reference to universals, top-level predicates such as sub-universal-of, instance-of, and $\uparrow$ -u-part-of remain crisp:

Axiom 8.2 (i) The universal a , referred to by the term Q_1 -*Georegion* on the precisification ω_1 , is a sub-universal-of b , the universal referred to by the term Q_2 -*Georegion* on the precisification ω_2 , if and only if every instance of a is an instance of b . (ii) a , the universal referred to by Q_1 -*Georegion* on the precisification ω_1 , is a $\uparrow$ -u-part-of b , the universal referred to by Q_2 -*Georegion* on the precisification ω_2 , if and only if every instance of a is an i-part-of some instance of b .

The term ω -*Georegion* is used to refer to a ω -*QGeoregion* universal without explicitly specifying the underlying quality pattern.

8.2. The Aristotelian method of classification applied to Georegions

The Aristotelian method of classification (Berg 1983) is based on the assumption that universals are structured hierarchically in a tree-like manner. Such trees are called the taxonomic trees or taxonomic hierarchies. Universals farther from the root in the sub-universal hierarchy are differentiated from universals closer to the root by additional quality universals. Those additional quality universals determine what marks out instances of a more specific universal (or species) within the wider parent universal (or genus) (Berg 1983). For example, in the definition '*Human* $\equiv$ *Rational Animal*' the quality universal *Rational* serves as differentia.

The Aristotelian method of classification is now used to identify the taxonomic structure of *Georegion* universals. Aside from the taxonomic hierarchy, this methodology also identifies hierarchical subdivisions of the surface of the Earth (or large i-parts of it). Formally, the notions of *taxonomic tree*, *partonomic tree*, and *subdivision tree* are distinguished (Bittner 2007):

Definition 8.3: Let $\Gamma(\omega) = (N, E)$ be a rooted acyclic graph such that the nodes in N refer to sub-universals of ω -*Georegion*. (i) $\Gamma(\omega)$ is a ω -*taxonomic tree* if and only if there is an edge $(n_1, n_2) \in E \subset N \times N$ if and only if the universal denoted by n_1 (on the precisification ω) is a sub-universal-of the universal denoted by n_2 (on the precisification ω). (ii) $\Gamma(\omega)$ is a ω -*partonomic tree* if and only if there is an edge $(n_1, n_2) \in E \subset N \times N$ if and only if the universal denoted by n_1 is a $\uparrow$ -u-part-of the universal denoted by n_2 . (iii) $\Gamma(\omega)$ is a ω -*subdivision tree* if and only if (a) the nodes in N denote ω -*Q-Georegion* universals, instances of which form ω -*Q*-partitions of x – the surface of the Earth (or large i-parts of it); and (b) there is an edge $(n_1, n_2) \in E \subset N \times N$ if and only if p_1 is a partition of x formed by a collection of instances of the universal denoted by n_1 , p_2 is a partition of x formed by a collection of instances of the universal denoted by n_2 ; and (c) p_1 is $\uparrow$ -p-included-in p_2 .

Taxonomic, partonomic, and subdivision trees for Bailey's system are discussed in Section 8.3. For more details see Bittner (2007).

Definition 8.4: $\Gamma(\Omega)$ is a Ω -*taxonomic tree array* (respectively, Ω -*partonomic tree array*, Ω -*subdivision tree array*) if and only if (a) for all $\omega \in \Omega$, $\Gamma(\Omega)[\omega] = \Gamma(\omega) = (N, E)$ is a

ω -taxonomic tree (respectively, ω -partonomic tree, ω -subdivision tree) and for all $\omega_1, \omega_2 \in \Omega$: $\Gamma(\Omega)[\omega_1]$ is isomorphic to $\Gamma(\Omega)[\omega_2]$.

The requirement of isomorphic graph structures in Ω -taxonomic (partonomic and subdivision) tree arrays indicates that the taxonomic, partonomic, and subdivision hierarchies, which are represented by those structures are not affected by the vagueness of the term *Georegion*. As required in Axiom 8.2, the predicates sub-universal-of, $\uparrow$ -u-part-of, and $\uparrow$ -p-included-in are crisp as assumed in the underlying top-level ontology.

Definition 8.5: $\Gamma(\Omega, \mathcal{Q}_B) = \langle \Gamma_t(\Omega), \Gamma_p(\Omega), \Gamma_s(\Omega) \rangle$ is an *Aristotelian georegion classification and delineation system* of the geographic region x if and only if:

- (1) $\Gamma_t(\Omega)$ is a Ω -taxonomic tree array and $\mathcal{Q}_B$ is a base genus pattern such that for all $\omega \in \Omega$: the nodes of $\Gamma_t(\Omega)[\omega]$ denote ω - $\mathcal{Q}'$ -Georegion universals such that (a) $\mathcal{Q}'$ is a sub-pattern of $\mathcal{Q}_B$ or (b) $\mathcal{Q}'$ is a species pattern of $\mathcal{Q}''$ which in turn is a sub-pattern of $\mathcal{Q}_B$. The edges of $\Gamma_t(\Omega)[\omega]$ correspond to genus-species relations between quality pattern of the respective ω - $\mathcal{Q}$ -Georegions.
- (2) $\Gamma_p(\Omega)$ is a partonomic tree array such that for all $\omega \in \Omega$: (a) the set N_p^ω of nodes of $\Gamma_p(\Omega)[\omega] = (N_p^\omega, E_p^\omega)$ is identical to the set N_t^ω of nodes of $\Gamma_t(\Omega)[\omega] = (N_t^\omega, E_t^\omega)$; and (b) if $(n_k, n_j) \in E_p^\omega$ (i.e., $n_k \uparrow$ -u-part-of n_j), then the quality pattern $\mathcal{Q}'$ that serves as a differentia in the definition of n_j (ω - $\mathcal{Q}'$ -Georegion) is a sub-pattern of the quality pattern $\mathcal{Q}$ that serves as differentia in the definition of n_k (ω - $\mathcal{Q}$ -Georegion).
- (3) $\Gamma_s(\Omega)$ is a subdivision tree array such that for all $\omega \in \Omega$: (a) all nodes of $\Gamma_s(\Omega)[\omega]$ denote ω - $\mathcal{Q}'$ -Georegions such that (i) $\mathcal{Q}' \subseteq \mathcal{Q}_B$ and (ii) instances of each denoted universal form a ω - $\mathcal{Q}'$ -partition of x and x is a geographic region of continental scale or larger; (b) one node of $\Gamma_s(\Omega)[\omega]$ denotes the universal ω - $\mathcal{Q}_B$ -Georegion.
- (4) The precisification range Ω is such that the underlying Ω - $\mathcal{Q}_B$ -partition is *maximally inhomogeneous*, that is, for all Ω' : if there is a Ω' - $\mathcal{Q}_B$ -partition of x , then $\Omega' \preceq \Omega$.

From Definition 8.5 (1–3) it follows: (a) the quality pattern $\mathcal{Q}_B$ is global in the partitions of x formed by the instances of ω - $\mathcal{Q}_B$ -Georegion; (b) the universal $\mathcal{Q}_B$ -Georegion is the leaf node of Γ_s . Definition 8.5(4) requires that Ω includes all ways of precisifying vague terms of the form ω - $\mathcal{Q}_B$ -Georegion that yield a reasonable and coherent delineation. That is, the vagueness of Ω - $\mathcal{Q}$ -Georegions is directly linked to the vagueness of the underlying Ω - $\mathcal{Q}$ -partitions. Consequently, the vagueness of the classification is directly linked to the vagueness of the delineation (and vice versa).

8.3. Bailey's ecoregion classification system

The ecoregion classification system of Bailey (1983) produces an Aristotelian classification and delineation system $\Gamma(\Omega, \mathcal{Q}-S) = \langle \Gamma_t(\Omega), \Gamma_p(\Omega), \Gamma_s(\Omega) \rangle$ of the *CUS* in the sense of Definition 8.5. This will be discussed for the taxonomic (Γ_t), the partonomic (Γ_p), and the subdivision hierarchy (Γ_s) in this subsection. As argued above, the predicates sub-universal-of, $\uparrow$ -u-part-of, and $\uparrow$ -p-included-in are crisp. Correspondingly the taxonomic graphs in Γ_t are all isomorphic and so are the partonomic graphs in Γ_p , and the subdivision graphs in Γ_s . For this reason it is sufficient to discuss Γ_t , Γ_p , and Γ_s for a single fixed precisification parameter. Here the precisification parameter ω_B that was used to draw the ecoregion maps published in the National Atlas (CEC 2009) and Figures 2 and 3 is chosen. In addition, it needs to be shown that the precisification range Ω underlying the classification and delineation is such that the Ω - $\mathcal{Q}_B$ -partition is maximally inhomogeneous as required in Definition 8.5 (4).

8.3.1. The taxonomic hierarchy

In Bailey's classification system the ecoregion universals are characterized in terms of maximal homogeneity with respect to the climate and land-surface quality pattern in Table 2. Land-surface qualities are integrated into the climate-based framework by forming the genus quality pattern Q - S that has climate quality pattern as well as land-surface quality pattern as sub-pattern. In addition, the quality universal *Elevation* (with *Lowland* and *Mountainous* as sub-universals) is included to capture the elevation dependency of climate quality pattern (Bailey 1983).

Definition 8.6: Universal c is a ω -Section (ω -Province, ω -Division, ω -Domain, respectively) if and only if every instance of c is a member of some collection that forms a ω - Q - S -partition (ω - Q - P , ω - Q - Di , and ω - Q - Do -partition with Q - P , Q - Di , and Q - Do , respectively, as defined in Table 2) of the surface of the Earth. All instances of ω -Section and ω -Province are, in addition, self-connected. Universal c is a ω -Ecoregion if and only if c is a ω -Section, ω -Province, ω -Division, or ω -Domain.

Thus, ω -Section (ω -Province, ω -Division, ω -Domain, respectively) is an abbreviation for ω - Q - S -Georegion (ω - Q - P -Georegion, ω - Q - Di -Georegion, ω - Q - Do -Georegion, respectively). That is, instances of ω -Section (ω -Province, ω -Division, ω -Domain, respectively) are geographic regions that are maximally ω -homogeneous with respect to SOME species of the genus quality pattern Q - S (Q - P , Q - Di , Q - Do , respectively). For example, every instance of ω -Domain is maximally ω -homogeneous with respect to some proper sub-universal-of *Climate regime*. The defining genus quality pattern of the universals ω -Domain, ω -Division, ω -Province, and ω -Section stand in the sub-pattern relation as depicted in Figure 4.

Definition 8.7: Universal c is a proper sub-universal of ω -Section (respectively ω -Province, ω -Division, ω -Domain) if and only if every instance x of c is an instance of ω -Section (ω -Province, ω -Division, ω -Domain, respectively) and x is ω -homogeneous with respect to ONE species of the genus quality pattern Q - S (respectively Q - P , Q - Di , Q - Do) as specified in Bailey (1983).

Parts of the resulting taxonomic hierarchy are depicted in Figure 5 for immediate sub-universals of ω -Domain (left) and for immediate sub-universals of ω -Division (right) in conjunction with the quality pattern that are involved in the definitions of the depicted ecoregion universals, as well as the genus-species relationships between those quality pattern. The immediate sub-universals of ω -Domain, ω -Division, and so on are the leaf

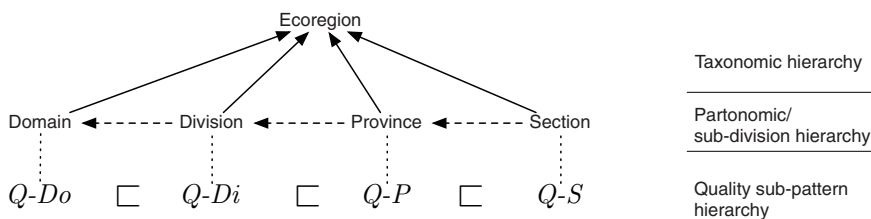


Figure 4. The ecoregion universals in Bailey's system. The solid arrows represent the relation sub-universal-of, the dashed arrows represent the relation $\uparrow$ -u-part-of, and $Q_k \sqsubset Q_j$ represents that Q_k is a sub-pattern of Q_j . A dotted line connecting a quality pattern to a universal indicates that this quality pattern is used as a differentia in the definition of the corresponding universal. (The ω -prefixes for *Ecoregion*, *Domain*, etc. are omitted.)

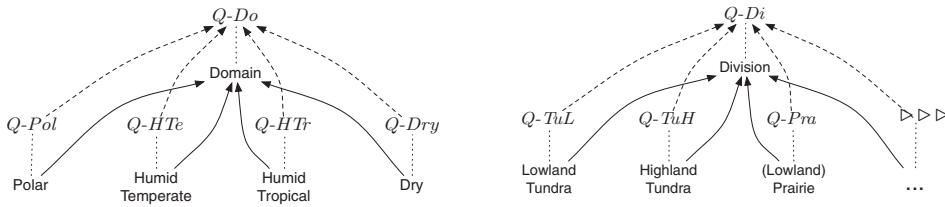


Figure 5. The *Domain* subgraph (left) and the *Division* subgraph (partly depicted: right) of the sub-universal-of hierarchy with genus and species pattern. The solid arrows represent the relation sub-universal-of, a dashed arrow between the quality pattern Q_k and Q_j indicates that Q_k is a species pattern of Q_j . A dotted line connecting a quality pattern to a universal indicates that this quality pattern is used as a differentia in the definition of the corresponding universal. The quality pattern $Q-Dry$, $Q-HTe$, $Q-HTr$, $Q-Pol$, $Q-TuL$, $Q-TuH$, and $Q-Pra$ were defined in Table 3.

nodes of the taxonomic tree. This rather flat taxonomic structure is an immediate consequence of the fact that all instances of ω -*Domain* (ω -*Division*, etc.) are members of ω - Q -*Do*-partitions (ω - Q -*Di*-partitions, etc.). (See (Bittner 2007) for details.)

By inspecting Figures 4 and 5, one can verify that the ecoregion classification system based on Definitions 8.6 and 8.7 produces the taxonomic component of an Aristotelian classification and delineation system in the sense of Definition 8.5 (1).

8.3.2. The partonomic hierarchy

The partonomic nesting of the universals ω -*Section*, ω -*Province*, ω -*Division*, and ω -*Domain* is depicted in Figure 4. The partonomic nesting of the proper sub-universals of such universals is partly depicted in the respective figures of (Bittner 2007). In addition, one can verify by inspecting Figure 2 and the ecoregion maps published in the National Atlas CEC (2009) that on the precisification $\omega_B \in \Omega$, the taxonomic and partonomic structures are related in the ways required in Definition 8.5 (2): if n_k is a $\uparrow$ -u-part-of n_j in the partonomy tree, then, in the corresponding taxonomy tree, the quality pattern that serves as a differentia in the definition of n_j is a sub-pattern of the quality pattern that serves as differentia in the definition of n_k . For example, $\uparrow$ -u-part-of holds between the universals ω_B -*Prairie Lowland Division* and ω_B -*Humid Temperate Domain*. The quality pattern serving as differentia in the definition of ω_B -*Prairie Lowland Division* is the pattern $Q-Pra$ (Table 2). The quality pattern serving as differentia in the definition of ω_B -*Humid Temperate Domain* is $\langle \text{Humid Temperate Climate Regime} \rangle$. Clearly, the latter is a sub-pattern of the former.

8.3.3. The subdivision hierarchy

To see that the conditions of Definition 8.5 which are related to the hierarchical sub-divisions (8.5 (1) and (3)) are satisfied in Bailey's system, consider Figure 4 and the precisification $\omega_B \in \Omega$. Instances of the immediate sub-universals of ω_B -*Ecoregion* (ω_B -*Domain*, ω_B -*Division*, ω_B -*Province*, and ω_B -*Section*), respectively, form ω_B - Q -*Do*-, ω_B - Q -*Di*-, ω_B - Q -*P*-, and ω_B - Q -*S*-partitions of the *CUS*. Map representations of these partitions can be found in Figure 2 and in Bailey (1983) and CEC (2009). As discussed in the examples for Definitions 5.5 and 6.5, those partitions are hierarchically nested in the sense that they are $\uparrow$ -p-included in one another. In addition, the quality pattern $Q-Do$, $Q-Di$, $Q-P$, and $Q-S$ are global in the respective partitions of *CUS* (see also Bittner (2007)).

8.3.4. Maximality of the range of admissible precisifications

As demonstrated in this section and documented in Bailey (1983, 1996, 1998), USFS (1995), and Cleland *et al.* (1997), Bailey's system is based on global (and thus nonlocal) quality pattern (including those listed in Tables 2 and 3). No additional local quality pattern are incorporated to decrease the degree of vagueness and to further precisify the delineation. Therefore none of the underlying Ω_B - $\mathcal{Q}$ -partitions is locally crisped. In particular:

Proposition 8.8: *The vague Ω_B - $\mathcal{Q}$ - S -partition of CUS is not locally crisped.*

From Lemma 7.4 it then follows:

Lemma 8.9: *The precisification range, Ω_B , of the vague Ω_B - $\mathcal{Q}$ - S -partition of CUS is such that the inhomogeneities permitted in this partition are maximal with respect to the quality pattern $\mathcal{Q}$ - S . In addition, the degree of vagueness of the vague Ω_B - $\mathcal{Q}$ - S -partition of CUS is maximal, that is, the set $\mathcal{P}_{(\mathcal{Q}-S, \Omega_B)}^{CUS}$ is maximal.*

The maximality of Ω_B in conjunction with the properties of the taxonomic, partonomic, and subdivision structures discussed in the previous subsections support the following corollary:

Corollary 8.10: *The ecoregion classification system of Bailey (1983) produces an Aristotelian classification and delineation system $\Gamma(\Omega, \mathcal{Q}-S) = \langle \Gamma_t, \Gamma_p, \Gamma_s \rangle$ of CUS in the sense of Definition 8.5.*

8.4. The trade-off between classification and delineation

If one accepts the assumptions specified in the axioms and definitions presented here, then one can prove the conjecture (1), which states that it is impossible to have both: a relatively precise delineation that incorporates local quality pattern by means of locally crisped partitions and a sophisticated Aristotelian classification based on nonlocal quality-based differentia. Using the vocabulary developed above, this can be stated as:

Lemma 8.11: *If $\Gamma(\Omega, \mathcal{Q}) = \langle \Gamma_t(\Omega), \Gamma_p(\Omega), \Gamma_s(\Omega) \rangle$ is an Aristotelian classification and delineation system for the region x , then the Ω - $\mathcal{Q}$ -partition of x is not locally crisped.*

Proof: Assume that $\Gamma(\Omega, \mathcal{Q}) = \langle \Gamma_t(\Omega), \Gamma_p(\Omega), \Gamma_s(\Omega) \rangle$ is an Aristotelian classification and delineation system for x . By Definition 8.5 (4), Ω is maximal. That is, for all Ω' : if there is a vague Ω' - $\mathcal{Q}$ -partition of x , then $\Omega' \preceq \Omega$. Assume that the Ω - $\mathcal{Q}$ -partition of x is a locally crisped version of the vague Ω' - $\mathcal{Q}$ -partition of x . By Definition 7.2, $\Omega \prec \Omega'$. That is, $\Omega' \not\preceq \Omega$. Because Ω is maximal it also follows that $\Omega' \preceq \Omega$. Thus $\Omega' \preceq \Omega$ and $\Omega' \not\preceq \Omega$. Hence, the Ω - $\mathcal{Q}$ -partition of x is not locally crisped. $\square$

Consider the system of the EPA. The increased precision of the delineation based on maximal homogeneity with respect to local quality pattern comes at a price: In the locally crisped delineation system it is impossible to identify *Georegion* universals using an Aristotelian classification system (Lemma 8.11). Instead of identifying universals, the EPA focusses on identifying and delineating ecoregion particulars. When displayed on a map those particulars form the collections *LIII*, *LII*, and *LI* displayed in Figure 1 with the properties discussed in Section 3.1.

9. Conclusions

The vagueness and granularity that characterize many classification and delineation of systems for geographic regions were analyzed in a mereology-based ontology. At the core

of this ontology is the notion of *homogeneity with respect to geographic qualities*. The vagueness and granularity of homogeneity give rise to a trade-off between the possibility to identify georegion universals using the Aristotelian method of classification and the preciseness of the delineation of georegion particulars. This trade-off is widely misunderstood and has led some to believe that the classification and delineation of geographic regions is more or less arbitrary or mostly policy based. By contrast, the analysis here has shown that it is logically coherent to believe that many kinds of geographic regions exist in reality and that they can be identified by scientific methods of classification and delineation.

However, because vagueness and granularity are unavoidable features of classification and delineation systems for many kinds geographic regions (particularly ecoregions, ecosystems, biomes, biotopes, etc.), the trade-off between classification and delineation poses a real challenge to the computational representation and processing of many geographic classification and delineation systems: The need for geometric representations, many of which rely on relatively precise boundaries, conflicts with the need for sophisticated classification systems for scientific and integration purposes. Understanding the true nature of the problem may be a first step toward overcoming it.

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